

TrES-3b

R-2675

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210611 · created 2026-08-30T22:48:00+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459376.6928 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1966 +/- 0.0084
Transit depth (Rp/Rs)^2	3.87 +/- 0.33 [%]
Orbital Inclination (inc)	82.84 +/- 0.44
Airmass coefficient 1 (a1)	1.023 +/- 0.012
Airmass coefficient 2 (a2)	-0.019 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#3 - [105, 310]
Optimal Aperture	3.18
Optimal Annulus	6.22
Transit Duration (day)	0.0653 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1966 ± 0.0084 vs 0.16309 (deviation 3.99σ)
Duration fit vs published	0.0653 d vs 0.0567 d (deviation 3.6σ)
Mid-time O–C	-4.6 min
Depth SNR	23.4
Residual scatter	1.15 %

Light curve

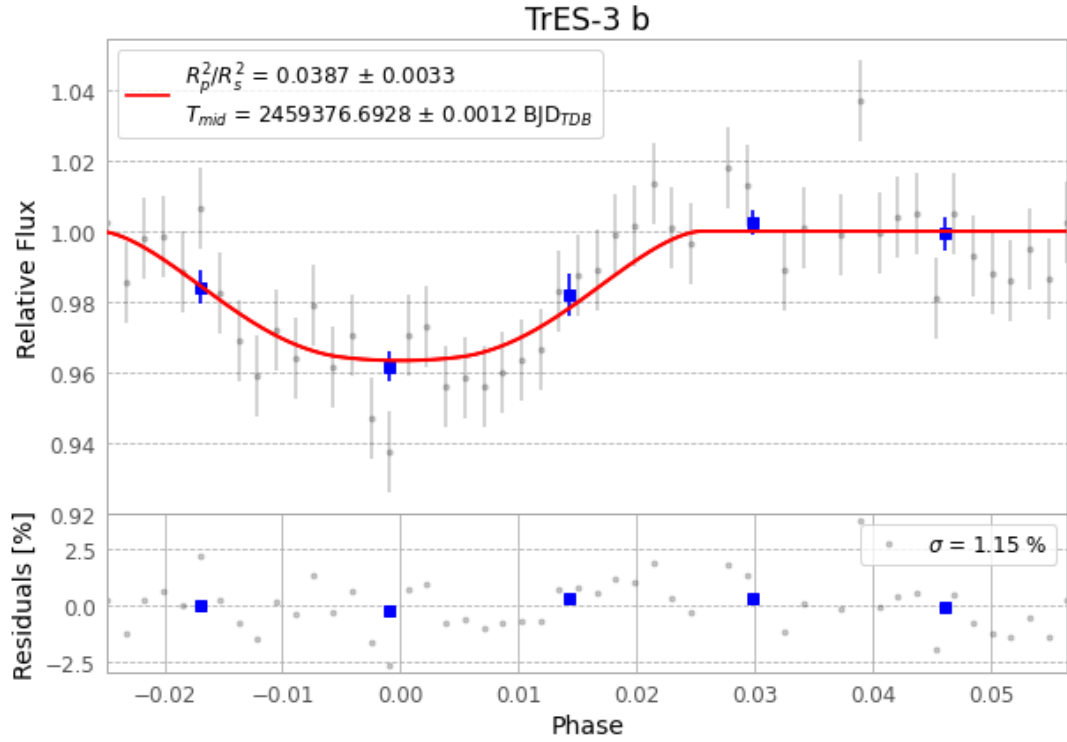


Figure 1 — FinalLightCurve_TrES-3 b_2021-06-10.png (SHA-256 a81be255790762de...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 69f5dc537cb5e4be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

52 FITS frames (34 MB), TRES-3210611034512.FITS → TRES-3210611061812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210611.log (bcd427b28aba...), FinalLightCurve_TrES-3 b_2021-06-10.png (a81be2557907...), FinalLightCurve_TrES-3 b_2021-06-10.csv (7eb0ce59c5ae...)

Compute resources

Wall time 148.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 22:48Z.